

Beliefs about Refugee Integration in Germany: A Survey Experiment on Ukrainian and Syrian Refugees

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Contents

1	Introduction	1
2	Literature Review and Conceptual Framework	1
2.1	Immigration Attitudes and Migrant Characteristics	1
2.2	Beliefs and Expectations about Migrants.	3
2.3	Differential Perceptions of Ukrainian and Syrian Migrants.	3
2.4	Contact, Personal Experience, Information, and Memory.	3
3	Experimental Design	3
3.1	Target Population and Proposed Sample	3
3.2	Factorial Vignette Design	3
3.3	Outcomes Measures	5
3.4	Respondent Questionnaire	5
3.5	Randomization and Survey Procedure	6
4	Empirical Strategy and Identification	6
5	Limitations and Extensions	7
5.1	Extensions	7
5.2	Limitations and Identification Constraints	7
6	Conclusion	7
	References	8

1 Introduction

2 Literature Review and Conceptual Framework

2.1 Immigration Attitudes and Migrant Characteristics

A large literature examines how host-country populations evaluate migrants and which characteristics shape these evaluations. One central distinction in this literature is based on whether attitudes toward migrants primarily reflect individuals' own economic interests and labor-market considerations or characteristics of the migrants.

Earlier economic studies provide some evidence that labor-market considerations are associated with immigration preferences (e.g., Scheve and Slaughter, 2001; Mayda, 2006). Other studies, however, emphasize the importance of welfare concerns, cultural considerations, and group-related attitudes rather than just individual labor-market interests (Burns and Gimpel, 2000; Dustmann and Preston, 2007; Sides and Citrin, 2007). Reviewing this literature, Hainmueller and Hopkins (2014) conclude that evidence for explanations based on individual economic self-interest is generally limited, particularly compared with broader sociotropic economic and cultural concerns.

Experimental evidence further challenges a simple labor-market competition explanation and rather points to the importance of migrants' own characteristics. Hainmueller and Hiscox (2010) find that both high- and low-skilled natives prefer high-skilled over low-skilled immigrants. If immigration attitudes were primarily driven by competition with similarly skilled workers, preferences would be expected to differ according to respondents' own skill levels. Instead, the common preference for high-skilled immigration suggests that respondents place considerable weight on migrants' human capital and potential economic contribution. This pattern is also found in multidimensional experiments. Higher education, occupational status, employability, and language proficiency are generally associated with more favorable evaluations of migrants (Hainmueller and Hopkins, 2015; Bansak et al., 2016; Czymara and Schmidt-Catran, 2016). More recent evidence from Germany similarly shows greater willingness to accept skilled labor migrants with recognized professional qualifications, good language skills, and a definite job offer (Wolff et al., 2025). Taken together, these studies show that characteristics associated with migrants' expected labor-market integration play an important role in how they are evaluated by the host-country population.

However, evaluations of migrants are not determined by human capital alone. Country of origin, cultural and religious background, and migration status also influence attitudes when other characteristics are held constant. Hainmueller and Hopkins (2015) find differences in evaluations across immigrants' countries of origin alongside effects of education, occupation, and English proficiency. In the European context, Bansak et al. (2016) show that preferences over asylum seekers depend jointly on economic, humanitarian, and cultural characteristics, with country of origin and religion affecting support in addition

to employability and vulnerability. Similar patterns emerge in Germany. Czymara and Schmidt-Catran (2016) find that acceptance varies with migrants' country of origin and migration motive as well as their education and German-language proficiency. These findings suggest that respondents evaluate migrants based on both individual characteristics and the broader groups and migration categories to which they belong. One possible explanation for the importance of such group-level characteristics is provided by theories of group position and perceived group threat. Quillian (1995), building on the group-position perspective of Blumer (1958), argues that prejudice toward immigrant groups can be related to perceived collective threat and the relative position of the in-group and out-group. In the European context, such perceptions are associated with factors including the relative size of immigrant groups and economic conditions. This perspective suggests that evaluations of migrants may partly reflect how migrant groups and their position within the host society are perceived.

Despite their differences, much of this literature shares a focus on attitudes and preferences toward migrants. Outcomes commonly concern support for immigration, willingness to admit or accept migrants, or preferences regarding their residence and employment. These outcomes capture how respondents evaluate migrants, but they do not necessarily reveal what respondents believe about migrants' likely outcomes. Attitudes and expectations are conceptually distinct. For example, a respondent may be willing to accept a refugee while expecting that the person will face difficulties finding employment or integrating socially. Conversely, a respondent may oppose further immigration while nevertheless expecting a particular migrant to succeed economically. Understanding how migrant characteristics affect beliefs about expected outcomes therefore requires examining these expectations directly, which is what I aim to do in this thesis.

This thesis focuses on what outcomes respondents expect different migrants to experience, rather than on whether they are willing to accept them. I examine expectations concerning employment, occupational match, future income, discrimination, and social integration. The proposed factorial design makes it possible to examine whether these expectations vary with migrant characteristics, while also testing whether otherwise comparable migrants are evaluated differently. The next section develops this distinction further by reviewing the literature on beliefs, expectations, and misperceptions about migrants.

2.2 Beliefs and Expectations about Migrants.

2.3 Differential Perceptions of Ukrainian and Syrian Migrants.

2.4 Contact, Personal Experience, Information, and Memory.

3 Experimental Design

The study employs a factorial vignette experiment in which respondents evaluate hypothetical migrants whose characteristics are independently varied across profiles. This approach is particularly suitable for the present research question because migrants differ simultaneously along several characteristics, such as country of origin, migration status, skill level, and language proficiency. Factorial designs allow the effects of these characteristics to be distinguished by independently varying them while holding the remaining characteristics constant. The proposed design builds on previous conjoint and factorial vignette experiments examining how migrant characteristics affect evaluations by host-country populations (Hainmueller and Hopkins, 2015; Bansak et al., 2016; Czymara and Schmidt-Catran, 2016).

- i indexes respondents.
- j indexes vignette profiles.
- $g \in \{\text{Ukraine, Syria}\}$ indexes migrant groups.
- Y_{ij} denotes respondent i 's evaluation of vignette j .
- C_{ig} denotes respondent i 's contact exposure to migrant group g .
- Q_{ig} denotes respondent i 's overall contact quality with migrant group g .
- M_{ig} denotes respondent i 's media exposure regarding migrant group g .

General Notation: For inline equations, use For displayed equations, use

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or the equation environment.

3.1 Target Population and Proposed Sample

3.2 Factorial Vignette Design

The study employs a factorial vignette experiment in which respondents evaluate hypothetical migrants whose characteristics are independently varied across profiles. This

approach builds on previous factorial and conjoint experiments examining how characteristics of immigrants and refugees affect evaluations by host-country populations (e.g., Hainmueller & Hopkins, 2015; Bansak et al., 2016; Czymara & Schmidt-Catran, 2016).

The vignettes describe potential migrants who currently live in Germany and differ randomly across several characteristics. The description of immigrants is based on a total of five dimensions: country of origin (Ukrainian/Syrian), reason for immigration (war refugee/labor migrant), gender (male/female), language skills (low/good German), and education (low/high). Country and language skills describe the cultural distance. Distinguish between two levels: (1) low and (2) good German language skills. The education factor represents the human capital of the immigrants. Distinguish between a (1) low and a (2) high level of education, where: low level of education: completed secondary education but has no vocational or university degree. high level of education: has a university degree. It should also be taken into account that language skills also have an economic component. Success in the labor market depends, for example, on the ability to speak the local language. The effect size of language proficiency could exceed the effect sizes of religious affiliation and country of origin. However, it should be considered that part of the effect of language skills, in the sense of the human capital hypothesis, can also be attributed to economic considerations. (Czymara & Schmidt-Catran, 2016)

Why religious affiliation is not included in the vignette: As discussed in the literature review, previous research demonstrates that religion and perceived cultural distance can influence attitudes toward immigrants (e.g., Czymara & Schmidt-Catran, 2016). However, in this study, religious affiliation is deliberately not included as a randomized vignette attribute. The present study focuses on differences in expectations associated with migrants' country of origin, migration status, human capital, gender, and German-language proficiency. Introducing religion as an additional vignette dimension would increase the complexity of the experimental design. Moreover, explicitly providing information about religious affiliation could alter the beliefs that respondents would otherwise associate with Ukrainian and Syrian migrants based on the country of origin alone. Religion is therefore left unspecified in the vignette, allowing the country-of-origin treatment to capture respondents' overall beliefs associated with the respective national-origin group. Consequently, the estimated country-of-origin effect should be interpreted as a composite effect that may encompass perceived cultural and religious differences, rather than as an effect of nationality that is independent of religion. Respondents' own religious affiliation is nevertheless collected as a background characteristic and can be included as a control or examined in heterogeneity analyses.

Vignette wording: [Gender: Mr./Mrs.] [random letter]. came from [country of origin: Ukraine/Syria] to Germany. [He/She] came to Germany as [reason for immigration: a refugee after fleeing the war in [his/her] country / a labor migrant seeking employment opportunities in Germany]. [He/She] [skill level: has completed a university degree / has no formal vocational qualification], speaks German [language skills: well / only to a

limited extent] and is currently looking for stable employment.

vignette universe: $2 \times 2 \times 2 \times 2 \times 2 = 32$ profiles

male/female $\times$ Ukraine/Syria $\times$ refugee/labor migrant $\times$ high/low skill $\times$ good/limited German = 16 profiles

Each respondent receives only a few randomly selected vignettes. The randomization allows estimation of the marginal effect of each characteristic.

1. Explained vignette attributes: *Origin_j MigrationStatus_j Gender_j Education_j GermanSkills_j*

2. Choosing particular vignette attributes: Educational attainment is included because previous vignette and conjoint evidence consistently documents more favorable evaluations of highly skilled immigrants (Hainmueller & Hiscox, 2010; Hainmueller & Hopkins, 2015; Czymara & Schmidt-Catran, 2016).L Language proficiency can be connected to Czymara & Schmidt-Catran and other relevant experiments. Migration status can be justified through the refugee-versus-labor-migrant literature. Country of origin obviously comes from your substantive Ukrainian–Syrian research question.

3.3 Outcomes Measures

3.4 Respondent Questionnaire

Citing measurement sources for the respondent questionnaire: C_{ig}^{Work} workplace contact frequency C_{ig}^{Social} social contact frequency Q_{ig} contact quality

Contact quality: The notation $Q_{ig} \in \{1, \dots, 7\}$ denotes respondent i 's overall contact quality with migrant group g . Contact quality is measured on a 7-point scale, where 1 indicates very negative contact experiences and 7 indicates very positive contact experiences. The measure is adapted to distinguish between the two origin groups examined in the present study.

$$PositiveExperience_{ig} = \max(Q_{ig} - 4, 0)$$

$$NegativeExperience_{ig} = \max(4 - Q_{ig}, 0)$$

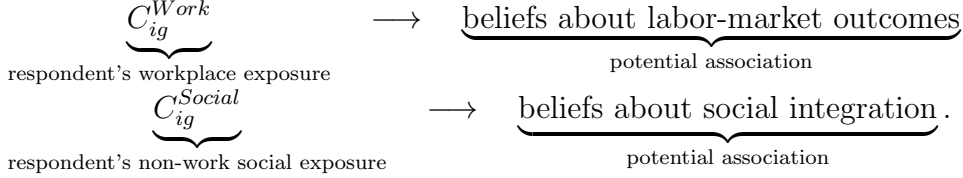
Following the intergroup-contact literature, the questionnaire distinguishes between the frequency and quality of respondents' contact with members of the two origin groups. This distinction is motivated by previous evidence showing that the quantity and valence of intergroup contact constitute conceptually distinct dimensions and may have different relationships with attitudes toward refugees (e.g., De Coninck et al., 2021). Contact is measured separately for Ukrainians and Syrians and across workplace and non-work social settings.

Media exposure: $MediaExposure_{ig}$ Media exposure is measured separately for Ukrainian and Syrian refugees. Building on work conceptualizing refugee-related media exposure as

a form of indirect intergroup contact (e.g., De Coninck et al., 2021), respondents report the frequency with which they encounter relevant news and media coverage. The measure is adapted to distinguish between the two origin groups examined in the present study.

Motivated by research showing that the memorability of information and narratives may influence subsequent beliefs (Graeber et al., 2024), the questionnaire additionally elicits whether respondents can recall a particularly salient personal experience involving members of either origin group and, if so, the valence of that experience.

Belief about migrant integration:



Beliefs about large communities and network effects:

Large community $\rightarrow$ More information/referrals $\rightarrow$ Better employment

versus

Large community $\rightarrow$ Less native interaction $\rightarrow$ Slower integration.

Construct:

$$\Delta B_i^{\text{Network}} = B_{i,S}^{\text{Network}} - B_{i,U}^{\text{Network}}$$

3.5 Randomization and Survey Procedure

4 Empirical Strategy and Identification

Identification and equations: Baseline equation:

$$Y_{ij} = \alpha + \beta_1 Ukr_j + \beta_2 Ref_j + \beta_3 Female_j + \beta_4 HighSkill_j + \beta_5 German_j + \varepsilon_{ij}.$$

Adjusted specification:

$$Y_{ij} = \alpha + \beta_1 Ukr_j + \beta_2 Ref_j + \beta_3 Female_j + \beta_4 HighSkill_j + \beta_5 German_j + \boldsymbol{\delta}' \mathbf{X}_i + \varepsilon_{ij}.$$

H4 regression with interaction term:

$$Y_{ij} = \alpha + \beta_1 Ukraine_j + \beta_2 Refugee_j + \beta_3 (Ukraine_j \times Refugee_j) + \boldsymbol{\gamma}' \mathbf{Z}_j + \varepsilon_{ij}.$$

where

$$\mathbf{Z}_j = \begin{pmatrix} Female_j \\ HighSkill_j \\ GoodGerman_j \end{pmatrix}$$

$$\mathbf{X}_i = (Age_i, Gender_i, Edu_i, Emp_i, Inc_i, Mig_i, East_i, Religion_i, Right_i)'.$$

Then,

$$Y_{ij} = \alpha + \beta' \mathbf{Z}_j + \boldsymbol{\theta}' \mathbf{X}_i + \varepsilon_{ij},$$

where $\boldsymbol{\theta}$ is the vector of coefficients associated with the respondent controls, and $\mathbf{Z}_j$ is contains the randomized vignette characteristics and $\mathbf{X}_i$ contains respondent-level controls.

5 Limitations and Extensions

5.1 Extensions

5.2 Limitations and Identification Constraints

6 Conclusion

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Statement of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Date:

Signature: